

# Lecture Notes on Thermodynamic Computing

## Chapter 6: Ultra-Low Voltage Circuit Design

### Chapter 6. Ultra-Low Voltage Circuit Design

#### 1 6.1 Introduction to Ultra-Low Voltage Operation

##### • Ultra-Low Voltage Regime Definition

**Definition 1.1** (Ultra-Low Voltage (ULV)). Circuit operation at supply voltages below or comparable to the thermal voltage:

$$V_{DD} \leq 3 - 5 \times V_T \approx 80 - 130 \text{ mV at } 300\text{K} \quad (1)$$

Key characteristics:

- Subthreshold transistor operation
- Exponential current-voltage relationships
- Dominant thermal noise effects
- Process variation sensitivity

##### • Energy Scaling Laws

Dynamic energy:

$$E_{dyn} = \frac{1}{2} C V_{DD}^2 \propto V_{DD}^2 \quad (2)$$

Subthreshold current:

$$I_{sub} = I_0 \exp\left(\frac{V_{GS} - V_{th}}{nV_T}\right) \quad (3)$$

Static energy:

$$E_{static} = V_{DD} \cdot I_{sub} \cdot t_{delay} \quad (4)$$

Delay scaling:

$$t_{delay} = \frac{C V_{DD}}{I_{sub}} \propto V_{DD} \exp\left(-\frac{V_{DD}}{nV_T}\right) \quad (5)$$

- **Energy-Delay Optimization**

Total energy per operation:

$$E_{total} = \frac{1}{2}CV_{DD}^2 + V_{DD} \cdot I_{leak} \cdot \frac{CV_{DD}}{I_{sub}} \quad (6)$$

Optimal supply voltage:

$$\frac{dE_{total}}{dV_{DD}} = 0 \Rightarrow V_{DD}^{opt} = nV_T \ln \left( \frac{C}{I_{leak} \cdot t_{target}} \right) \quad (7)$$

Minimum energy point (MEP):

$$V_{MEP} \approx 2.5 - 3 \times nV_T \quad (8)$$

## 2 6.2 Subthreshold Circuit Design Principles

- **Transistor Sizing in Subthreshold**

Transconductance efficiency:

$$\frac{g_m}{I_D} = \frac{1}{nV_T} \quad (9)$$

This is independent of transistor size, unlike strong inversion.

Optimal sizing strategy:

- Minimize capacitance for speed
- Increase width for drive strength
- Balance leakage vs performance

- **Logic Gate Design**

Subthreshold inverter:

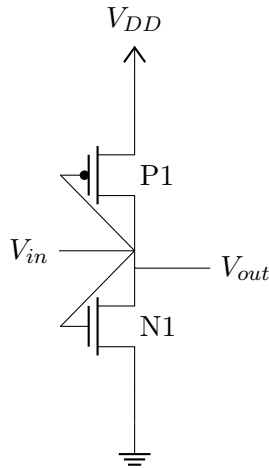


Figure 1: Subthreshold CMOS inverter

Switching threshold:

$$V_M = \frac{nV_T}{1 + \sqrt{\frac{\mu_n W_n}{\mu_p W_p}}} \ln \left( \sqrt{\frac{\mu_n W_n}{\mu_p W_p}} \right) + \frac{V_{DD} + V_{th,p} - V_{th,n}}{2} \quad (10)$$

Noise margins:

$$NM_H = V_{DD} - V_{IH}, \quad NM_L = V_{IL} \quad (11)$$

### • Complex Gates

NAND gate analysis:

$$I_{out} = \begin{cases} I_{p1} & \text{if } A = 0 \text{ or } B = 0 \\ \min(I_{n1}, I_{n2}) & \text{if } A = 1 \text{ and } B = 1 \end{cases} \quad (12)$$

Stack effect in subthreshold:

$$I_{stack} = I_0 \exp \left( \frac{V_{GS1}}{nV_T} \right) \exp \left( \frac{V_{GS2} - \eta V_{DS1}}{nV_T} \right) \quad (13)$$

where  $\eta$  is body effect parameter.

## 3 6.3 Process Variation Effects

### • Threshold Voltage Variations

Pelgrom model:

$$\sigma_{V_{th}} = \frac{A_{V_{th}}}{\sqrt{WL}} \quad (14)$$

Typical values:  $A_{V_{th}} = 5 - 20 \text{ mV} \cdot \text{m}$

Current variation in subthreshold:

$$\frac{\sigma_{I_D}}{I_D} = \frac{\sigma_{V_{th}}}{nV_T} \quad (15)$$

For  $\sigma_{V_{th}} = 50 \text{ mV}$  at 300K:

$$\frac{\sigma_{I_D}}{I_D} = \frac{50 \times 10^{-3}}{1.3 \times 25.9 \times 10^{-3}} = 1.48 = 148\% \quad (16)$$

### • Statistical Design Methods

Monte Carlo analysis for yield optimization:

$$P_{failure} = P \left( \frac{t_{delay}}{t_{spec}} > 1 \right) \quad (17)$$

Design centering:

$$\mu_{design} = \mu_{spec} - k\sigma_{spec} \quad (18)$$

where  $k$  depends on target yield (e.g.,  $k = 3$  for 99.7

- **Adaptive Body Biasing**

Body bias voltage control:

$$V_{th}(V_{BS}) = V_{th0} + \gamma(\sqrt{|\phi_B + V_{BS}|} - \sqrt{|\phi_B|}) \quad (19)$$

Compensation strategy:

$$V_{BS} = -K \times (V_{th,measured} - V_{th,target}) \quad (20)$$

Power overhead:

$$P_{ABB} = I_{junction} \times V_{BS} + P_{control} \quad (21)$$

## 4 6.4 Temperature Effects and Compensation

- **Temperature Dependence**

Threshold voltage temperature coefficient:

$$\frac{dV_{th}}{dT} = -1.5 \text{ to } -2.5 \text{ mV/K} \quad (22)$$

Mobility temperature dependence:

$$\mu(T) = \mu(T_0) \left( \frac{T}{T_0} \right)^{-\alpha} \quad (23)$$

where  $\alpha \approx 1.5 - 2.0$ .

Subthreshold current vs temperature:

$$\frac{1}{I_D} \frac{dI_D}{dT} = \frac{1}{nV_T} \left( \frac{dV_{GS}}{dT} - \frac{dV_{th}}{dT} \right) - \frac{1}{T} \quad (24)$$

- **Zero Temperature Coefficient Design**

Condition for temperature-independent current:

$$\frac{dI_D}{dT} = 0 \Rightarrow \frac{dV_{GS}}{dT} = \frac{dV_{th}}{dT} - nV_T/T \quad (25)$$

PTAT voltage generation:

$$V_{PTAT} = \frac{k_B T}{q} \ln(r) \quad (26)$$

where  $r$  is current density ratio.

- **Bandgap Reference for ULV**

Sub-1V bandgap reference:

$$V_{ref} = V_{BE} + \alpha V_{PTAT} \quad (27)$$

For zero temperature coefficient:

$$\alpha = -\frac{dV_{BE}/dT}{dV_{PTAT}/dT} \approx 22 \quad (28)$$

Startup circuit requirements:

- Overcome degenerate bias point
- Ensure proper PTAT operation
- Minimize power consumption

## 5 6.5 Noise Analysis in ULV Circuits

### • Thermal Noise

Input-referred thermal noise:

$$\overline{v_n^2} = \frac{4k_B T \gamma}{g_m} \quad (29)$$

In subthreshold:  $\gamma = 1/2$  (vs  $\gamma = 2/3$  in strong inversion)

SNR limitation:

$$SNR = \frac{V_{signal}^2}{\overline{v_n^2}} = \frac{V_{signal}^2 g_m}{2k_B T} \quad (30)$$

### • Flicker Noise

$1/f$  noise in subthreshold:

$$\overline{v_{1/f}^2} = \frac{K_f}{WLC_{ox}f} \quad (31)$$

Noise corner frequency:

$$f_c = \frac{K_f g_m^2}{4k_B T \gamma WLC_{ox}} \quad (32)$$

ULV circuits have higher  $f_c$  due to reduced  $g_m$ .

### • Noise-Optimized Design

Minimum noise figure amplifier:

$$NF_{min} = 1 + 2\sqrt{\frac{\gamma k_B T}{g_m R_s}} \quad (33)$$

Optimum source resistance:

$$R_{s,opt} = \sqrt{\frac{\gamma k_B T}{g_m}} \quad (34)$$

## 6 6.6 Memory Circuits in ULV

### • ULV SRAM Design

6T SRAM cell stability:

$$SNM = \min(SNM_H, SNM_L) \quad (35)$$

Read stability metric:

$$RSNM = \frac{V_{DD}}{2} - \sqrt{\frac{I_{access}}{I_{cell}}} \times \frac{V_{DD}}{2} \quad (36)$$

Write margin:

$$WM = V_{DD} - V_{trip} \quad (37)$$

- **Assist Techniques**

Read assist using negative bitline:

$$V_{BL} = -V_{assist} < 0 \quad (38)$$

Write assist using supply boosting:

$$V_{DD,boost} = V_{DD} + \Delta V_{boost} \quad (39)$$

Dynamic voltage scaling during access:

$$V_{DD}(t) = V_{DD,standby} + \Delta V \cdot \text{Access\_Enable}(t) \quad (40)$$

- **Data Retention Voltage**

Minimum retention voltage:

$$V_{DD,min} = \max(V_{hold,0}, V_{hold,1}) \quad (41)$$

Hold SNM calculation:

$$HSNM = V_{trip} - \frac{V_{DD}}{2} \quad (42)$$

For reliable retention:  $HSNM > 3\sigma_{noise}$

## 7 6.7 Clock and Timing in ULV

- **Clock Generation**

Ring oscillator frequency:

$$f_{osc} = \frac{1}{2N\tau_d} = \frac{I_D}{2NCV_{DD}} \quad (43)$$

where  $N$  is number of stages and  $\tau_d$  is delay per stage.

Temperature compensation:

$$f(T) = f_0 \left( \frac{T}{T_0} \right)^\alpha \exp \left( \frac{E_a}{k_B} \left( \frac{1}{T_0} - \frac{1}{T} \right) \right) \quad (44)$$

- **Clock Distribution**

Clock skew in ULV:

$$t_{skew} = \frac{C_{load}\Delta V_{th}}{g_m V_{DD}} \quad (45)$$

H-tree distribution:

$$P_{clock} = f_{clk} \times C_{total} \times V_{DD}^2 \quad (46)$$

Clock gating efficiency:

$$\eta_{gating} = 1 - \frac{P_{leakage} + P_{gating\_overhead}}{P_{dynamic}} \quad (47)$$

- **Timing Closure**

Setup time in subthreshold:

$$t_{setup} = \frac{C_{node}}{g_m} \ln \left( \frac{V_{DD}}{V_{min}} \right) \quad (48)$$

Hold time consideration:

$$t_{hold} = t_{clk\_delay} - t_{data\_delay} \quad (49)$$

Timing uncertainty:

$$t_{uncertainty} = t_{jitter} + t_{skew} + t_{margin} \quad (50)$$

## 8 6.8 Power Management Techniques

- **Dynamic Voltage and Frequency Scaling**

Frequency-voltage relationship:

$$f_{max} \propto \frac{(V_{DD} - V_{th})^\alpha}{V_{DD}} \quad (51)$$

Energy optimization:

$$E = \frac{CV_{DD}^2 f_{target}}{f_{max}(V_{DD})} \quad (52)$$

Optimal voltage:

$$\frac{dE}{dV_{DD}} = 0 \quad (53)$$

- **Power Gating**

Sleep transistor sizing:

$$W_{sleep} = \frac{I_{active}}{I_{unit}} \times K_{margin} \quad (54)$$

Wake-up time:

$$t_{wakeup} = \frac{C_{virtual} V_{DD}}{I_{sleep}} \quad (55)$$

Leakage reduction:

$$I_{sleep} = I_{active} \times \exp \left( -\frac{V_{sleep}}{nV_T} \right) \quad (56)$$

- **Adaptive Supply Voltage**

Droop detector:

$$V_{droop} = L \frac{di}{dt} + R \Delta i \quad (57)$$

Voltage regulation:

$$V_{DD}(t) = V_{target} - K_p e(t) - K_i \int e(t) dt \quad (58)$$

where  $e(t) = V_{target} - V_{measured}(t)$ .

## 9 Examples

**Example 9.1** (Optimal ULV Supply Voltage). Find optimal supply voltage for minimum energy per operation.

**Given:**

- Load capacitance:  $C = 10$  fF
- Leakage current:  $I_0 = 1$  pA
- Subthreshold slope:  $n = 1.3$
- Target delay:  $t_{target} = 1$  s
- Temperature:  $T = 300$  K

**Solution:**

Thermal voltage:

$$V_T = \frac{k_B T}{q} = 25.9 \text{ mV} \quad (59)$$

Total energy:

$$E_{total} = \frac{1}{2} C V_{DD}^2 + V_{DD} I_0 t_{target} \exp\left(-\frac{V_{DD}}{n V_T}\right) \quad (60)$$

Taking derivative and setting to zero:

$$\frac{dE}{dV_{DD}} = C V_{DD} + I_0 t_{target} \exp\left(-\frac{V_{DD}}{n V_T}\right) \left(1 - \frac{V_{DD}}{n V_T}\right) = 0 \quad (61)$$

Solving numerically with given values:

$$V_{DD}^{opt} \approx 3 \times n V_T = 3 \times 1.3 \times 25.9 = 101 \text{ mV} \quad (62)$$

Minimum energy:

$$E_{min} = \frac{1}{2} \times 10 \times 10^{-15} \times (0.101)^2 \quad (63)$$

$$+ 0.101 \times 10^{-12} \times 10^{-6} \times \exp(-3) \quad (64)$$

$$= 51 \text{ aJ} + 5 \text{ aJ} = 56 \text{ aJ} \quad (65)$$

**Example 9.2** (Process Variation Impact). Calculate yield for ULV logic gate with threshold voltage variations.

**Given:**

- Supply voltage:  $V_{DD} = 150$  mV
- Nominal threshold:  $V_{th} = 400$  mV
- Threshold variation:  $\sigma_{V_{th}} = 50$  mV
- Gate width:  $W = 120$  nm, Length:  $L = 65$  nm



- Required current:  $I_{min} = 1$  pA

**Solution:**

Subthreshold current:

$$I_D = I_0 \frac{W}{L} \exp\left(\frac{V_{DD} - V_{th}}{nV_T}\right) \quad (66)$$

Current variation:

$$\frac{\sigma_{I_D}}{I_D} = \frac{\sigma_{V_{th}}}{nV_T} = \frac{50}{1.3 \times 25.9} = 1.48 \quad (67)$$

Log-normal distribution parameters:

$$\mu_{ln} = \ln(I_{nominal}) - \frac{1}{2} \ln(1 + 1.48^2) \quad (68)$$

$$\sigma_{ln} = \sqrt{\ln(1 + 1.48^2)} = 1.31 \quad (69)$$

Probability of meeting specification:

$$P(I_D > I_{min}) = 1 - \Phi\left(\frac{\ln(I_{min}) - \mu_{ln}}{\sigma_{ln}}\right) \quad (70)$$

For typical values: Yield 85

## 10 Homework Problems

### Assignment 6.1: Ultra-Low Voltage Circuit Design

#### 1. Energy Optimization:

- Find MEP for given technology parameters
- Compare dynamic vs static energy contributions
- Design voltage regulator for optimal point

#### 2. Process Variation Analysis:

- Calculate current mismatch in differential pair
- Design offset compensation circuit
- Perform Monte Carlo yield analysis

#### 3. Temperature Compensation:

- Design PTAT current generator
- Implement zero-TC bias circuit
- Calculate temperature drift over -40°C to 85°C

#### 4. Noise Analysis:

- Calculate SNR for ULV amplifier
- Design noise-optimal input stage

- Compare thermal vs flicker noise

**5. SRAM Design:**

- Design 6T cell for 100 mV operation
- Calculate read/write margins
- Implement assist circuits

**6. Clock Systems:**

- Design temperature-compensated oscillator
- Calculate clock skew in H-tree
- Optimize clock gating strategy

**7. Power Management:**

- Implement DVFS controller
- Design power gating switches
- Calculate wake-up energy overhead

**8. System Integration:**

- Design complete ULV processor
- Optimize power-performance trade-offs
- Implement adaptive voltage scaling

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**Next Chapter Preview:** Chapter 7 will explore "Thermodynamic Processors and CPUs" including complete processor architectures operating in the thermodynamic regime.